

Improvement of representative days selection in power system planning by incorporating the extreme days of the net load to take account of the variability and intermittency of renewable resources

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HIGHLIGHTS

- A way to include the extreme days of the net load in representative days.
- The suitable number of representative days is found by two simple methods.
- Dispatchable capacity is underestimated if the extreme days are not included.
- The importance of assigning proper weights to the extreme days.
- Several years of the net load data are used to select representative days.

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ABSTRACT

As the share of variable renewable energy in power systems increases, incorporating the operational constraints in generation expansion planning has become necessary to address flexibility issues. Accordingly, to preserve chronology and to make the technically detailed planning model tractable, the model should be solved for some representative periods. Different approaches have been introduced to select representative periods; however, most of them ignore the importance of the net load. In this research, a method is proposed to include the extreme days with higher and lower levels of the net load in representative days by using self-organizing map clustering. Further, the impacts of the extreme days, with different weighting approaches, on planning and operational decisions are analyzed for single-year and multi-year planning. A simple method is also devised to find the proper number of representative days. These methods are applied to the NREL-118 and the Belgian power systems by deploying a planning model based on the unit commitment problem. It was shown that investment costs and the required dispatchable capacity are greatly underestimated if the extreme days of the net load are not considered. Besides, using self-organizing map to weight the extreme days according to their importance in the historical data resulted in 45% improvement in approximating operating costs, a better estimation of operating variables, and a more reliable expansion for multi-year planning, demonstrating the significance of assigning proper weights to extreme days. The method for the number of representative days also performed well, especially when the extreme days were included.

1. Introduction

Deployment of renewable resources should substantially grow in the coming years in order to decarbonize power systems, and achieving this goal mainly depends on investing heavily in wind and solar power [1]. As a tool for evaluating the integration of such energy sources in power systems, optimization models called generation expansion planning (GEP) are widely used for finding technically realistic solutions [2].

However, the intermittency and variability of variable renewable energy (VRE), such as wind and solar photovoltaic (PV) power, increase the need for operational flexibility, posing problems in GEP.

Since taking some actions to improve power system flexibility, such as investing in more flexible power plants, takes several years, it is essential to consider flexibility in GEP when VRE comprises a high share of power generation [3,4]. Not considering flexibility issues could also lead to a capacity portfolio that is unable to satisfy the net load

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Nomenclature*Acronyms*

GEP	Generation expansion planning
O&M	Operation and maintenance
VRE	Variable renewable energy
DC	Duration curve
NLDC	Net load duration curve
SOM	The self-organizing map
DTW	Dynamic time warping
CC	Combined cycle power plants
CT	Combustion turbine power plants
ST	Steam turbine power plants
NU	Nuclear power plants
Bio	Biomass power plants
NG	Natural gas
PV	Photovoltaic
Opt	The basic, hybrid optimization model to select representative days
Ward	Ward's clustering method
PAM	The partitioning around medoids algorithm

Sets

$t \in T$	All generation units (technologies)
$t \in T^{\text{New}}$	New units
$t \in T^{\text{Ex}}$	Existing units
$t \in T^{\text{D}}$	Dispatchable units
$t \in T^{\text{ND}}$	Non-dispatchable units
$t \in T^{\text{DR}}$	Dispatchable units that can participate in providing operating reserve
$t \in T^{\text{RE}}$	Renewable power generation units
$y \in Y$	Years
$d \in D$	Representative days
$h \in H$	Hours
$i \in I$	Interconnection lines
$p \in P$	Time series of a power system

Variables

$\text{cap}_{a,y,t}$	Available capacity of unit t in year y (MW)
$\text{pg}_{y,t,d,h}$	Power generated by each unit t at hour h of representative day d in year y (MW)
$\text{imp}_{y,i,d,h}$	Power imported through interconnector i at hour h of representative day d in year y (MW)
$\text{exp}_{y,i,d,h}$	Power exported through interconnector i at hour h of representative day d in year y (MW)
$\text{res}_{\text{dr}_{y,t,d,h}}$	Downward regulating reserve provided by a dispatchable unit $t \in T^{\text{D}}$ at hour h of representative day d in year y (MW)
$\text{res}_{\text{ur}_{y,t,d,h}}$	Upward regulating reserve provided by a dispatchable unit $t \in T^{\text{D}}$ at hour h of representative day d in year y (MW)

$\text{res}_{\text{us}_{y,t,d,h}}$ Upward spinning reserve provided by a dispatchable unit $t \in T^{\text{D}}$ at hour h of representative day d in year y (MW)

Binary variables

$\text{inv}_{y,t}$	Investment decision. Equals 1 if a new unit $t \in T^{\text{New}}$ is constructed in year y
$\text{stu}_{y,t,d,h}$	Start-up decision. Equals 1 if a dispatchable unit $t \in T^{\text{D}}$ starts up at hour h of representative day d in year y .
$\text{cmt}_{y,t,d,h}$	Commitment status of dispatchable units $t \in T^{\text{D}}$. Equals 1 if a dispatchable unit $t \in T^{\text{D}}$ is committed and operational at hour h of representative day d in year y .
$\text{std}_{y,t,d,h}$	Shutdown decision. Equals 1 if a dispatchable unit $t \in T^{\text{D}}$ shuts down at hour h of representative day d in year y .

Parameters

DR_t	Discount rate of each unit (p.u.)
DRI	Discount rate of interconnection lines (p.u.)
BY	Base year
$\text{Cost}_{V_{y,t}}$	Capital costs of unit t in year y (\$/MW)
$\text{Cost}_{F_{y,t}}$	Variable costs of power generation, including variable O&M and fuel costs, of unit t in year y (\$/MWh)
$\text{Cost}_{F_{y,t}}$	Fixed O&M costs of unit t in year y (\$/MW)
$\text{Cost}_{\text{Stu}_t}$	Start-up costs of each dispatchable unit $t \in T^{\text{D}}$ (\$)
Cost_{I_y}	Costs of power import in each year (\$/MWh)
Price_{E_y}	Price of power export in each year (\$/MWh)
RMR	Reserve margin ratio (p.u.)
Cap_t	Capacity of each unit (MW)
$\text{Cap}_{E_{y,t}}$	Capacity of existing units in each year (MW)
$\text{Cap}_{\text{Fct}_{t,d,h}}$	Capacity factor of each non-dispatchable units $t \in T^{\text{ND}}$ at each hour of each representative day (p.u.)
FOR_t	Forced outage rate of dispatchable units $t \in T^{\text{D}}$ (p.u.)
RepW_d	Weight of each representative day (p.u.)
Lifetime_t	Lifetime of each unit (year)
$\text{PD}_{y,d,h}$	Power demand at hour h of representative day d in year y (MW)
MSG_t	Minimum stable generation level of dispatchable units $t \in T^{\text{D}}$ (p.u.)
MDT_t	Minimum down time of dispatchable units $t \in T^{\text{D}}$ (hour)
MUT_t	Minimum up time of dispatchable units $t \in T^{\text{D}}$ (hour)
RDL_t	Ramp down limit of dispatchable units $t \in T^{\text{D}}$ (MW/hour)
RUL_t	Ramp up limit of dispatchable units $t \in T^{\text{D}}$ (MW/hour)
Res_{URF}	The Fraction of hourly load that is equal to the system upward regulating reserve requirement
Res_{DRF}	The Fraction of hourly load that is equal to the system downward regulating reserve requirement
Res_{USF}	The Fraction of hourly load that is equal to the system upward contingency spinning reserve requirement
IC_{Imp_i}	Interconnection capacity of line $i \in I$ for power import
IC_{Exp_i}	Interconnection capacity of line $i \in I$ for power export
MSRE_y	The minimum share of demand supplied by renewable resources in each year

variability in the future; hence, it is cost-effective to address these issues in planning for power systems [5]. It has also been shown that including operational flexibility constraints in long-term planning models produces more realistic results for systems with high penetration of VRE [6], and determines more flexible capacity such as natural gas combined cycle (CC) units [7]. On the other hand, overestimation of the potential of VRE and baseload generators besides underestimation of operating costs are all consequences of a stylized representation of power system technical details in a planning model [8]. As a result, if

constraints capturing dynamics of power system operation are not included in GEP, there would be a huge discrepancy in costs and performance of the optimal generation technology mix when the share of VRE in electricity generation is significant. Many researchers, thus, have incorporated unit commitment constraints, such as start-up and operating reserve constraints, into GEP [9].

To model operational constraints such as minimum down and up time, the temporal representation of the planning model must preserve chronology. It is also shown that lack of retaining chronology in time

slices used in GEP leads to underestimation of gas-fired power plants and overestimation of VRE capacity [10]. Nonetheless, it is impossible to run a detailed planning model with a high temporal resolution for the whole of several decades. A number of representative periods with an hourly resolution are therefore used to assure tractability and to retain chronology. By using representative periods and consequently capturing the variability of load and VRE generation endogenously, operational flexibility of a power system can be assessed [11]. The most common representative period that has been investigated in the literature is representative days, also called typical days. Despite the above-mentioned advantages of representative days, selecting a set of days that holds the characteristics of the entire historical period is demanding [8].

Representative periods can be chosen manually. For instance, Koltsaklis and Dagoumas [12], opted for eight representative days corresponding to workweeks and weekends of four seasons in a year to enhance technical aspects of an optimization-based framework for the European electricity markets. In another work [13], twelve days representing each month of a year were randomly chosen for capturing seasonal fluctuations of demand and VRE to be used in a stochastic planning model of uncertain power markets. This simplistic approach to selecting representative days does not guarantee that a subset of days preserving the main characteristics of load and VRE time series is selected [8]. Besides, it is very likely this approach does not take account of both rare and common days appropriately because the weights of the manually selected days could not reflect the importance and repetition of these days in the historical data [11]. These are the reasons why various methods for choosing representative periods have been introduced and utilized in the literature, namely hierarchical clustering, such as Ward's clustering [14], partitional clustering, such as k-medoids [15], and k-means [16], clustering, and an optimization model [17].

Choosing the types of time series for selecting representative periods depends on the energy system aimed to be analyzed. In district energy systems, for instance, Fazlollahi et al. [18], selected representative days by clustering profiles of solar irradiation, electricity price, heating and electricity demand. Heijde et al. [19], also introduced an optimization model to conserve chronology between representative days selected from the mentioned profiles besides the temperature and solar thermal collector output to model seasonal thermal energy storage. In power system planning and operation, however, representative periods are mostly selected by both load and VRE time series. For example, load, wind, and solar generation of each day were combined to create the input vectors for the selection of representative days by a new hybrid clustering [20], by k-medoids clustering to be used in a model that establishes continuity between the selected days [15], by k-means clustering to investigate the system value of various technologies in the future UK power system [21], and by k-means clustering to reduce the computational cost of a large-scale GEP for a real case [22]. In another study [23], hourly load and VRE were clustered by k-means to reduce the computational cost and capture the operational dynamics for a generation and transmission co-optimization model in a system with the presence of electric vehicles. Pineda and Morales [24], also employed load, wind and solar data of each hour as the input vectors to propose a clustering method that preserves the chronology between elements of the obtained clusters. Indeed, each instance of a data set used in clustering methods consists of load and VRE of the historical period [14], and the error to approximate the historical load and VRE duration curves (DCs) is minimized in the optimization method [17], but there are some exceptions.

Sun et al. [25], employed hierarchical clustering on the objective domain, consisting of investment costs of each technology, to select representative days. Teichgraber and Brandt [26], used electricity price data to select representative days since they investigated operation problems of battery charge/discharge optimization and gas turbine scheduling. Taking power import price into account besides load and VRE, Heuberger et al. [27], employed representative hours to

investigate the effects of technology cost reduction on GEP. In few studies, the net load (residual load or net demand) is used instead of load and VRE data to select days. For instance, de Sisternes and Webster [28], introduced a way to select a subset of weeks that more accurately approximates the net load duration curve (NLDC), but their proposal is not applicable to shorter periods since the selected weeks were found by testing all the possible subsets. Green et al. [16], performed an analysis of power system operation by clustering load and net load separately, and they reported that more representative profiles had been needed to adequately represent the net load due to its higher variability. Scott et al. [29], used the net load implicitly by fixing the days with maximum and minimum net load in a clustering algorithm, in which load and VRE time series are clustered to find representative days. The net load curve and its fluctuations do not have a major role in selecting representative days in their proposed method since load and VRE curves are the only input data for clustering. However, a way to explicitly take advantage of the net load along with load and VRE data has not been devised to select representative days for power system planning, to the authors' knowledge. We presume that representative days should retain the main characteristics of not only load and VRE time series but also the net load. Thus, all of these curves must individually have an active role in selecting representative days. This idea is important because the net load demonstrates the amount of load that must be satisfied by dispatchable units, and its extreme periods would have a decisive effect on power systems planning. Therefore, we propose a method in which each of the net load besides load and VRE time series individually affects representative days selection. In selecting representative days, we focused on the extreme periods, which is explained in the next paragraph.

Some authors attempted to take account of extreme periods. The definition of extreme periods depends on the energy system under consideration. As an example of a residential supply system, Kotzur et al. [30], defined extreme days as the days with peak heat load, peak electricity load and the smallest photovoltaic feed-in. In other works done for district energy systems [18,31], the periods during which the peak of heating and cooling demand occurs were also deemed to be the extreme periods. A few studies took extreme periods into account for power system planning. Pfenninger [32], regarded days with the maximum and minimum of average wind generation, solar PV generation and power demand as extreme days and add them to the selected days for power system planning. The drawback of this definition is that actual weight of the extreme days is unknown. Extra computation burden is also imposed due to the attachment of up to 6 days to the other representative days. The extra, unnecessary computation burden due to the inclusion of many extreme days can also be seen in Frew and Jacobson's work [33], in which a way was introduced to find 8 representative days. More importantly, it should have been declared what definition is used to describe extreme days in [33]. It is important since the procedure for selecting extreme days stems from the definition used. It was also not declared how the weights of these periods are calculated in this research. Scott et al. [29], specified various types of extreme days, such as days with the highest increase or decrease of the net load over consecutive periods. They have shown that the extreme days do not necessarily exist in representative days obtained from clustering methods. It was also demonstrated that using extreme days in planning models leads to a more accurate estimation of the system costs. The definitions they used for extreme periods are interesting and valuable, but there are some difficulties regarding the methodology developed. The weighting of the representative days is one of the problems that should be improved according to their suggestion for future works. The weighting method developed in the study relies on the future net load and cannot accurately model the endogenous expansion of VRE. Besides, the modified k-medoids algorithm they introduced could be problematic because some instances are forced to be medoids of some of the clusters throughout the algorithm. As a result, the outputs of the algorithm are far different from when the standard

procedure for partitioning clustering is followed. The extreme days are found manually in the works relating to extreme periods in GEP, so there could be errors in selecting extreme days especially when the number of them is more significant in a power system with high penetration of VRE.

Since power system planning is the scope of this research, extreme periods are defined as the periods with the highest and lowest values of the net load. These critical periods determine the required dispatchable capacity for satisfying the load that is not met by VRE technologies, so they are crucial in power system planning and capacity expansion. In the proposed methodology, we strived to address the issues discussed above concerning the selection of extreme days in power system planning. By deploying the self-organizing map (SOM) clustering on the net load time series, a method for finding and separating the extreme days of the net load is devised. The proposed method is then attached to approaches to selecting the normal representative days out of load and VRE curves. As a result, the net load besides load and VRE time series individually have a key role in finding representative days.

Another advantage of the proposed method over the mentioned methods in the literature is that only two extreme days are added to the other representative days, causing a minor computation burden. Additionally, methods for selecting the normal days, such as k-medoids and ward's algorithms and the optimization method, work in a normal manner, with no disturbance. More importantly, the weighting of the extreme days is done endogenously by the SOM in a way that it reflects the importance of the extreme days according to the number of similar days in the historical data. The importance of proper weighting of the extreme days are further elaborated in the discussion sections, and it was demonstrated that assigning proper weights to the extreme days have a significant impact on investment and operational decisions. Allowing for scaling power generation and operating costs of a representative day to an annual amount, the weights of representative days are an essential part of representative days selection. Days holding similar load and VRE characteristics should be represented by the same chosen day. In this regard, the weight of a representative day should be equal to the number of its similar days in a year, so the weights indicate how important each representative day is in the historical data. Weighting of the normal days, i.e. the days that are not considered the extreme days of the net load, is straightforward. In clustering techniques the number of instances in a cluster indicates the corresponding weight of a normal day, and in the optimization method [17], weights are determined by the optimization model. However, we focus on weighting the extreme days of the net load and show its importance. Another merit of the introduced method is that the extreme days are automatically chosen. By the increase in the share of VRE generation in power systems, the number of the extreme days of the net load would become greater, so this feature of the proposed method becomes much more significant in this regard. The proposed approach to finding representative days is used for planning for a single year and multiple years. The effect of employing the extreme days on power system planning for multiple years has not been studied in the literature, to the author's knowledge. Besides, the underlying reasons for the improvements obtained by the inclusion of the extreme days are discussed in this work, which is lacking in the literature. Thus, we investigate how the investment and operational decisions are affected when the extreme days of the net load, with different weighting approaches, are included in representative days.

Furthermore, it is essential to use multiple years of historical data to select representative days because the inter-annually variability of VRE cannot be captured by data of a single year [32]. In few studies [14,16,22], VRE and load data of multiple years were used to select representative days. Hilbers et al. [34], also introduced a method that can be applied to multiple decades for finding timesteps with higher importance, i.e. greater variable costs, for capacity expansion. The net load is employed in this study to extract the extreme days; however, the mentioned works in the literature have not suggested a way to use

multiple years of the net load. The levels of the net load depend on the installed VRE capacity and can dramatically change over years by the addition of wind and solar PV units, so it is impossible to compare these values to find the extreme days. This problem is not encountered when only load and VRE time series, i.e. capacity factor of VRE units, are deployed for representative days selection. Thus, a solution is offered in this study to make the levels of the net load comparable and preserve the relationship between load and VRE generation over years.

In addition, there is not a general consensus about the proper number of representative days. Pfenninger [32], employed the elbow method to discover the suitable number of clusters. In this method, the total intra-cluster distance is plotted against the number of clusters. Fazlollahi et al. [18], took advantage of clustering validation measures, namely the average intra-cluster and inter-cluster distances. Solving the planning model for 1 to 100 representative days, Nahmmacher et al. [14], identified the sufficient number of days that can accurately estimate the total system cost. They also observed that more accurate results were obtained by applying more representative days. de Sisternes and Webster [28], determined the number of representative weeks by comparing the error between the real NLDC and its approximation from the selected weeks. In this research, we aim to introduce a simple, new method for choosing the number of representative periods by applying the metrics introduced in the literature for approximating the system DCs. The proposed method does not rely on the time-consuming process of planning model execution for different number of representative days. Clustering validation measures, moreover, are not included in our approach because they are not able to assess the effect of the number of representative days on approximating load and VRE patterns. Teichgraber and Brandt also showed that better clustering measures do not necessarily result in a better approximation of the objective value of a power system optimization problem [26]. Further, these measures cannot be used for non-clustering approaches to selecting representative days, so we offer a method that fits all the approaches.

Deploying the extreme values of the net load to improve the quality of representative days is the main focus of the present work. Some solutions are also offered to meet other challenges regarding temporal representation of energy planning models, which includes choosing the suitable number of representative days and using several years of historical data. Hence, the contributions of this research are categorized as follows:

- 1) By employing the SOM clustering, a method is proposed to find and separate the extreme days of the net load. This method can be attached to any approach to selecting the other representative days, which is done in this work for the clustering and optimization approaches. The superiority of the proposal is that the weights of the extreme days are endogenously specified according to the number of similar days, which reflects the importance of these days in the historical data. The extreme days are also selected automatically, which becomes more critical by the increase of VRE generation. Further, the chosen representative days have the essential characteristics of each of the net load, load and VRE time series because all of these curves are directly included in the procedure for selecting representative days.
- 2) By applying a planning model to two test systems, it is shown how the investment and operational decisions are affected by the inclusion of the extreme days of the net load, and the underlying reasons are discussed. The effect of using the extreme days, with different weighting approaches, is also investigated for both planning for a single year and multiple years. Generally, including the extreme days led to a much more reliable capacity expansion, and fine weighting of the extreme days by the SOM clustering brings about a more accurate estimation of operational and planning decisions.
- 3) It is problematic to employ multiple years of historical net load data to select the extreme days because the levels of the net load depend on the installed VRE capacity and considerably change over years.

Thus, a way is introduced to make the levels of the net load comparable so that multiple years of historical net load can be employed.

4) A simple way is suggested to choose the number of representative

days by applying the metrics introduced in the literature for approximating the DCs of power systems. The results show that the suggested method finds a suitable figure for the number of representative days.

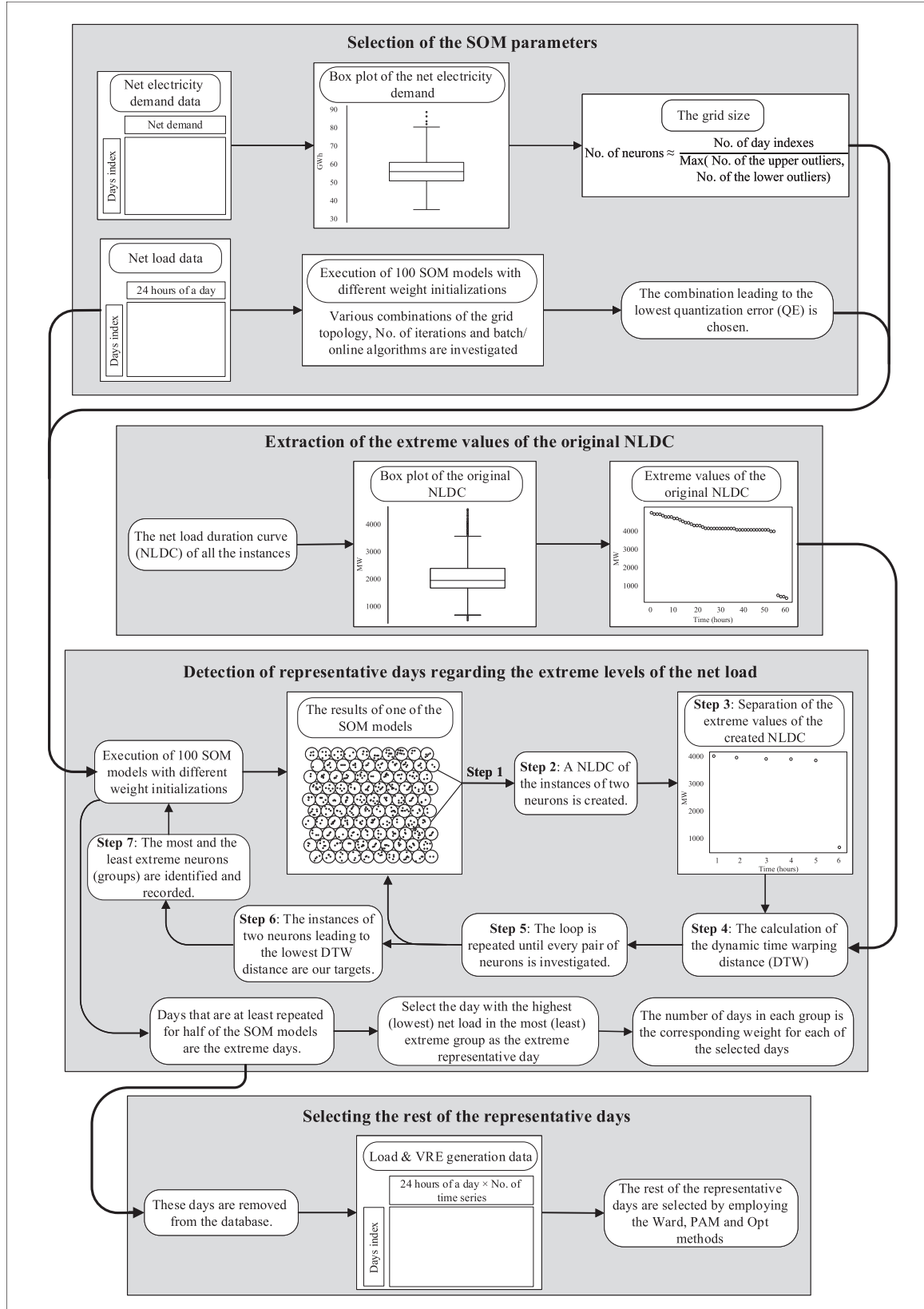


Fig. 1. A schematic of the proposed approach to incorporating the extreme days of the net load.

The remainder of this paper is as follows. The new ways to select representative days and the number of them are proposed in [Section 2](#). The GEP model deployed in case studies is introduced in [Section 3](#). [Section 4](#) explains the test systems and results obtained. Conclusions are presented in [Section 5](#).

2. Methodology

2.1. The proposed approach to incorporating the net load

Days with extreme net load values are critical in terms of capital and operating expenses in power system planning. In the case of low VRE generation and high load, more dispatchable capacity is required to meet the demand, giving rise to a huger investment in generation capacity. By contrast, when generation from VRE is high and demand is low, some dispatchable units must shut down or some VRE sources should be curtailed to guarantee the system security, causing a jump in system operational costs. These are the reasons why the extreme days of the net load are the focus of this study. The net load time series, on the other hand, cannot portray the characteristics of its essential components, i.e. load and VRE generation curves. In fact, each time series has its unique features that are not necessarily exhibited by the net load, such as daily pattern of load or zero generation of solar PV during the night. As a result, it is sensible to take each time series into account, besides the net load, in order to select representative days.

A method consisting of two steps is put forward in the work to choose representative days. Firstly, two representative days pertaining to the least and the most extreme levels of the net load are selected by using only the net load curve. Secondly, the selected extreme days of the net load are set aside, and the rest of representative days, i.e. normal days, are then chosen by taking advantage of load and VRE generation time series. The SOM clustering is employed for the first step, and approaches explained in [Section 2.1.3](#) are used for the second one. One would argue that days during which the peak and trough of the net load have occurred can be used for the first step, instead of clustering the net load. Although two extreme days are found easier by this approach, appropriate weights are not assigned to these extreme days. In fact, the number of similar days to those with the minimum and maximum net load is not clear, especially when the share of VRE is higher in a power system. However, the difference between these two approaches to extracting and weighting the extreme days are elaborated in the discussion sections.

2.1.1. The self-organizing map (SOM)

The SOM is a clustering method developed by Kohonen [35]. It is constructed by a two-dimensional lattice, called a map or grid, containing a number of neurons (nodes) arranged in a rectangular or hexagonal topology. The objective of the SOM is to map the input space to the grid in a way that similar instances are placed together in a neuron. The weight vector of each neuron is randomly initialized, and a procedure is followed to assign input vectors to neurons at each iteration. To do so, the Euclidean distances between a sample vector from the input space and every weight vector are computed. Then, a neuron whose weight vector is the closest to the sample vector, called best matching unit (BMU), is detected. The weight vectors of the BMU and its neighbors are subsequently updated to better reflect the features of the sample vector. The described procedure is known as online learning. A variant of the algorithm is batch learning, in which identifying the BMUs and updating of weight vectors are simultaneously done for all the instances in the input space. Both algorithms terminate when repeated for a predefined number of iterations.

2.1.2. Extracting the extreme days

The performance of SOM relies on many parameters, including the initialization of weight vectors, map size, number of algorithm iterations, grid topology and the choice between batch or online algorithms.

The quantization error (QE) is used to determine the number of iterations, type of grid topology, and the use of either online or batch algorithm in this study. The QE is defined as the mean Euclidean distances between instances (vectors) represented by each neuron and the weight vector of the neuron, which are then averaged out for all the neurons. However, the QE is not useful for deciding on the grid size [36]; therefore, we suggest a particular solution to obtain a figure for the map size for the specific problem of extracting the extreme days. The step-by-step process to tune the mentioned parameters, corresponding to the segment “selection of SOM parameters” in [Fig. 1](#), is presented as follows:

- Firstly, the size of the SOM grid should be determined. The idea behind our suggestion is that there should be sufficient neurons in the grid so that each neuron can on average hold as many instances as the number of extreme days of the net electricity demand. The net electricity demand is defined as the sum of the net load over hours of each day, i.e. the total net demand of electricity in each day, and the number of its extreme values can be obtained through box plot. Accordingly, the box plot of the daily net electricity demand should be created to display the number of its outliers, i.e. values greater than the sum of 1.5 times interquartile range (IQR) and the upper quartile or smaller than the lower quartile minus 1.5 times IQR. The approximate number of neurons is then figured by dividing the total number of instances by the maximum value between the number of outliers concerning the most extreme and the least extreme days. It is advisable that the dimensions of two lattices on the map be as close as possible to each other for constructing the sufficient number of neurons.
- After the grid size is found, the number of algorithm iterations, type of grid topology, and the choice between batch or online learning can be tuned by the QE. It is suggested that various combinations of the three parameters be created, and multiple SOM models, for instance 100 models, with random weight initializations be executed for each of the parameter combinations by using the net load time series. The parameter combination resulting in the lowest QE can be employed for SOM modeling [37]. It is noteworthy that the random weight vectors used for all of the parameter combinations should be the same to nullify the effect of the weight initialization on parameter tuning.

The way to neutralize the impact of the initial weights on extracting the extreme days is put forward at the end of this subsection. The SOM model is run after parameters are specified. Only the net load time series is applied for clustering by the SOM, and it is divided into instances (vectors) in a way that each instance represents each day while the features of the input space are hours of days. The features should be scaled by standardizing the data.

The idea behind our approach to extracting the extreme days of the net load is that a neuron in the SOM grid pertains to the days with the highest net load levels, and another neuron represents days with the lowest values of the net load. It is a valid assumption since days with a similar net load pattern are grouped together in a neuron. Thus, two neurons in the grid are sought that can jointly portray the extreme values of the net load duration curve (NLDC). The extreme values of the NLDC could be obtained by using the box plot in the same way it has been suggested earlier. Indeed, values of the NLDC that are beyond the whiskers are considered as outliers or the extreme levels of the net load. To find two neurons that can more exactly reflect the extreme values of the NLDC, the following steps should be taken after the execution of the SOM clustering:

- 1- Two neurons are selected from the SOM grid.
- 2- The instances of the selected neurons are grouped together, and a NLDC concerning these instances is created.
- 3- The values of the created NLDC that are greater than the upper

- whisker or are smaller than the lower whisker of the original NLDC are separated. These figures are the extreme levels of the created NLDC.
- 4- The dynamic time warping (DTW) distance between the extreme values of the original NLDC and those of the created NLDC is computed. If the created NLDC has no extreme values, the distance is deemed infinite.
 - 5- The aforementioned steps are implemented for every pair of neurons in the grid.
 - 6- The instances of two neurons leading to the lowest DTW distance are options for representing the least and the most extreme levels of the original NLDC.
 - 7- It is easy to identify the neuron relating to the most extreme and the least extreme levels of the net load by checking instances of the two neurons. However, the values of their elements can help us to avoid the manual search. The neuron holding instances with the higher average of the net load pertains to the most extreme net load levels, and vice versa.

Since the initialization of the weight vectors affects the outcome of the SOM modeling, several SOM models with different weight initializations, for instance 100 models, should be executed and followed by the steps mentioned above. The day indexes of the most and the least extreme days are separately recorded after each implementation of the SOM model and the steps. When the implementation of the 100 SOM clustering is done, two groups of day indexes pertaining to the most and the least extreme levels of the net load are available. These two sets of day indexes are respectively called E^M and E^L , where I^M and I^L are respectively the obtained day indexes of the options for the most and least extreme days. The number of repetitions of these days after the implementation of SOM and the steps is defined by n . The indexes of the two sets are shown by i and j .

$$E^M = \{(I_1^M, n_1^M), (I_2^M, n_2^M), \dots, (I_i^M, n_i^M)\} \quad (1)$$

$$E^L = \{(I_1^L, n_1^L), (I_2^L, n_2^L), \dots, (I_j^L, n_j^L)\} \quad (2)$$

Because of the random weight initialization of SOM clustering, some of these recorded days are repeated so rarely that it would be better not to consider them as the extreme days. Thus, the recorded days of the two groups that are repeated more than a predefined threshold are regarded as the extreme days of the net load. We used a threshold of 50, meaning the days that are at least repeated for half of the SOM models are the extreme days. Thus, if a day index has $n_i^M \geq 50$ for the most extreme days or $n_j^L \geq 50$ for the least extreme days, the day is considered as the final extreme day, and the other days are withdrawn. Accordingly, new sets $\tilde{E}^M$ and $\tilde{E}^L$ are defined, where $\tilde{I}$ is the index of the remaining days, and p and q are new indexes of the two new sets.

$$\tilde{E}^M = \{(\tilde{I}_1^M, \tilde{n}_1^M), (\tilde{I}_2^M, \tilde{n}_2^M), \dots, (\tilde{I}_p^M, \tilde{n}_p^M)\} \quad (3)$$

$$\tilde{E}^L = \{(\tilde{I}_1^L, \tilde{n}_1^L), (\tilde{I}_2^L, \tilde{n}_2^L), \dots, (\tilde{I}_q^L, \tilde{n}_q^L)\} \quad (4)$$

Then, the day with the highest net load in the most extreme group and the day with the lowest net load in the least extreme group are regarded as the chosen extreme representative days that will be used in GEP. Since the days in each of the final groups are the most similar ones to the selected extreme day of the corresponding group, the number of days in the most (least) extreme group determines the weight of the most (least) extreme representative day. Accordingly, the weight of the most extreme representative days is $|\tilde{E}^M| = P$, and the weight of the least extreme representative day is $|\tilde{E}^L| = Q$.

The DTW is an algorithm measuring similarity between two time series. Despite the Euclidean distance measure, the DTW is an elastic similarity measure, which means that each data point of a time series is compared to all the points in the other one to compute the DTW distance. Indeed, the DTW distance measure seeks to find an alignment

between two time series, called warping path, in a way that the sum of the absolute difference between the aligned points is minimized. Since the points of two time series can be aligned in numerous ways, some restrictions are imposed to limit the search space of warping paths. For example, the first and final points of warping paths must be the alignment of the initial and last points of two time series respectively. Movement in the warping paths is also restricted to the neighboring points [38]. Time series do not have to be equal in length for the DTW distance calculation, which is the main reason for using this similarity measure in the step 4.

2.1.3. Selecting the rest of the representative days

When representative days are used for temporal details in GEP, operating variables should be scaled to annual values by the weights of representative days. Thus, the sum of the weights must be equal to the number of days in a year, for the correct values are calculated for yearly variables. Accordingly, after the detection of days with extreme net load levels, these days should be removed from the database. In this regard, the other representative days are selected without involving the extreme days so that the sum of the weights of all the selected days, including the normal and extreme ones, exactly equals the number of days in the initial database. For instance, if selecting representative days of a year with 365 days is pursued, the sum of the weights of the normal days equals $365 - P - Q$, in which P and Q are respectively the weights for the most and least extreme representative days.

If several years of data are used, the same action should be taken. However, the acquired weights of all representative days must be divided by the number of years because the sum of the weights have the same value as the number of days in a year.

Load and VRE generation time series are then used to select the rest of the representative days by deploying three different methods, including (1) Ward's hierarchical clustering (Ward), (2) k-medoids clustering (PAM) and (3) the optimization model (Opt). These methods are preferred since they are the most sophisticated or widely-used techniques in the literature.

The optimization method has been proposed by Poncelet et al. [17]. This method uses a mixed integer linear programming (MILP) optimization problem to find the best subset of days and their corresponding weights. However, using integer variables to select the optimal subset of days leads to intensive computation and longer solution time. The authors, thus, have suggested a hybrid approach in which randomly selected days are fixed in the MILP model, so only the weights are optimized. This approach, called the basic hybrid optimization model (Opt), is employed in this study. Ward's hierarchical clustering [39], algorithm was first used by Nahmmacher et al. [14], to select representative days in power system planning. K-medoids clustering is preferred to k-means in this study. Medoid is defined as a time series in a cluster whose sum of Euclidean distances to the other elements in the cluster is minimal. The reason why k-medoid is picked is that choosing medoids as representative days results in a more realistic planning and better approximation of DCs compared to the case that the mean value of clusters is chosen [14,32]. Another merit of employing medoids is that these instances are among the real historical data of a power system and have the actual load and VRE generation characteristics. The partitioning around medoids (PAM) algorithm [40], is used to implement k-medoids clustering in this study. Despite k-means, the PAM algorithm produces the same clustering results for each execution in most cases.

For clustering by the PAM and Ward's algorithms, the data are structured in a way that each vector represents each day. The number of features also equals the number of time series multiplied by 24 h a day; i.e. time series of load and VRE generation are placed alongside each other. Further, the features should be scaled by standardizing the data before clustering. A schematic diagram of the overall approach proposed in order to incorporate the net load in selecting representative days is displayed in Fig. 1.

2.1.4. The application to multi-regional power systems

The proposed method described in the previous subsections is suitable for selecting representative days in a power system with a single region, but it can be used for a multi-regional GEP with some minor modifications. Since power systems used as case studies in this work are assumed to have one region for simplicity, the proposed method in the previous sections is implemented in this study. However, we offer three ideas to adjust the method for multi-regional systems.

The way to select the normal representative days in the second step of these three suggestions is straightforward, for its application has been shown in the literature for the Ward, PAM, and Opt methods. For instance, each vector of the input data set for clustering approaches should consist of daily load and VRE data for all regions. The first step for extracting the extreme days can be adapted by the three suggestions as follows:

- All regions can be considered together, and the data of all regions should be aggregated to form the net load of the entire system. The way to extract the extreme days is the same as before, and they are selected from the obtained net load. The final extreme days should be removed from the database for executing the second step.
- The second idea is that the net load of all regions participates in selecting the extreme days of the entire system. Three adjustments relating to data sets have to be made. Firstly, the features of the data used for SOM clustering should consist of regions in a way that each vector holds the net load of all regions. In this regard, SOM clustering groups days by considering the net load of all regions. Secondly, the hourly net load of all regions has to be sorted to create the NLDCs in order to find the extreme values of the net load subsequently. Thirdly, it is required that the sum of hourly net load in all regions be used to calculate the net electricity demand of each day to find the grid size by box plot. The rest of the procedure does not change.
- The extreme days of the net load in each region are separately found by the same method introduced for a single region, followed by deleting the extreme days of all regions from the database to find the normal days. Hence, two extreme representative days are used in GEP for each region. This idea is not very useful for a power system with many regions since the number of the extreme days employed in GEP and the computational cost of the planning model would become too high.

2.2. Ways to choose the number of representative days

Poncelet et al. [17], have proposed and gathered some metrics that measure the accuracy of approximating the real time series, i.e. load and VRE generation, by extending the time series of representative days. These metrics include the average normalized root mean square error (NRMSE) of approximating DCs ($NRMSE_{av}^{DC}$), the average correlation error (CE_{av}) and the average NRMSE of the approximation of ramp duration curves ($NRMSE_{av}^{RDC}$).

$NRMSE_{av}^{DC}$ computes how the distribution of each actual DC is preserved by its approximation $\tilde{DC}$. To construct each $\tilde{DC}_p$, values of time series p for representative days are sorted while each selected day is extended according to its weight. $NRMSE_{av}^{DC}$ is formulated as follows:

$$NRMSE_{av}^{DC} = \frac{\sum_p \left(\frac{\frac{1}{\sqrt{\|H\|}} \cdot \sum_h (DC_{p,h} - \tilde{DC}_{p,h})^2}{\max(DC_p) - \min(DC_p)} \right)}{\|P\|} \quad (5)$$

CE_{av} represents whether the correlation between real time series is maintained by the time series of representative days. To quantify this criterion, the average of the absolute difference between the Pearson correlation of each two real time series $corr_{p_i,p_j}$ and that of the corresponding time series for representative days $c\tilde{orr}_{p_i,p_j}$ is calculated as follows:

$$CE_{av} = \frac{2}{\|P\| \cdot (\|P\| - 1)} \cdot \left(\sum_{p_i} \sum_{p_j > i} |corr_{p_i,p_j} - c\tilde{orr}_{p_i,p_j}| \right) \quad (6)$$

$NRMSE_{av}^{RDC}$ measures the distribution of short-term fluctuations. The ramp duration curve (RDC) of each time series is generated by differentiating and sorting the time series afterwards.

$$NRMSE_{av}^{RDC} = \frac{\sum_p \left(\frac{\frac{1}{\sqrt{\|H\|}} \cdot \sum_h (RDC_{p,h} - \tilde{RDC}_{p,h})^2}{\max(RDC_p) - \min(RDC_p)} \right)}{\|P\|} \quad (7)$$

In this research, the above-mentioned metrics are calculated for different number of representative days. Regarding the fact that lower values of each metric mean a better approximation of the actual time series, the suitable number of representative days can be found in two ways:

- 1- Rigorous manner: The number of representative days leading to the lowest value for each metric is determined. The highest figure, among three options for the number of representative days, could be considered as the proper number of representative days.
- 2- Acceptable manner: Since higher number of representative days may be chosen by the first method, resulting in a lengthy execution of the planning model, another similar way is suggested. For each metric, the first figure for the number of days that its metric value is lower than the minimum of the metric plus its standard deviation is discovered. The highest figure, among three options for the number of representative days, could be regarded as the accepted number of representative days.

The application of both manners is displayed by illustrative examples in Section 4.1.2.

3. Generation expansion planning formulation

The unit commitment problem forms the basis of the GEP developed in this study. Thus, the impact of VRE generation intermittency and variability on the system investment decisions is endogenously included in the model. To capture operational flexibility, renewable energy curtailment, operating reserve and power import and export are considered. The model is suitable for a single region.

For the sake of simplicity, the GEP used in this study is a deterministic model applicable to a vertically integrated power industry with a monopolistic energy market. However, stochastic programming can be used for future works to handle the uncertainties of some parameters such as load and VRE generation to investigate the effects of the extreme days more thoroughly. Representative periods for reflecting temporal details could also be used in a deregulated market to simulate the competition between generator companies by other techniques such as multi-level programming.

3.1. Objective function

Total discounted system cost is minimized to determine the optimal capacity mix that can satisfy the system constraints. Specifically, the total discounted cost consists of: (a) discounted capital investment, (b) discounted fixed Operation and maintenance (O&M) costs, (c) discounted variable operating costs of dispatchable units (sum of variable O&M, fuel and start-up costs), (d) discounted variable operating costs of non-dispatchable units and (e) discounted cost of net import over the planning horizon. The objective function formulation is derived from the OSeMOSYS [41], energy modeling.

Start-up costs of dispatchable units are neglected in the first hour of each representative day since these days are modeled separately in each year. Indeed, most dispatchable units continue to operate in reality,

while the mathematical model determines that these units must start up at the beginning of representative days to satisfy load. This hypothesis was confirmed by observing much higher errors in estimating variable operating costs when start-up costs of all hours were included compared to not taking account of them for the first hours.

Since operating variables, such as costs and generation from units, are determined for each representative day in each year, the amount of these variables in representative days should be multiplied by the corresponding weights to calculate the yearly values in terms (c) to (e). Accordingly, the sum of the weights of representative days must be equal to the number of days in a year. Due to the role of the weights in calculating variable operating costs in the objective function, the weights have a significant effect on the optimal solution.

$$\begin{aligned} \text{MinCost} = & \sum_{y,t \in T^{\text{New}}} \frac{\text{Cost}_{C,y,t} \cdot \text{Cap}_t \cdot \text{inv}_{y,t}}{(1 + \text{DR}_t)^{(Y-BY)}} + \sum_{y,t} \frac{\text{Cost}_{F,y,t} \cdot \text{cap}_{a,y,t}}{(1 + \text{DR}_t)^{(Y-BY+0.5)}} \\ & + \sum_{y,t \in T^D} \frac{\sum_d \text{RepW}_d \cdot (\sum_h \text{Cost}_{V,y,t} \cdot \text{pg}_{y,t,d,h} + \sum_{h>1} \text{Cost}_{\text{Stu}_t} \cdot \text{stu}_{y,t,d,h})}{(1 + \text{DR}_t)^{(Y-BY+0.5)}} \\ & + \sum_{y,t \in T^{\text{ND}}} \frac{\sum_d \text{RepW}_d \cdot \sum_h \text{Cost}_{V,y,t} \cdot \text{pg}_{y,t,d,h}}{(1 + \text{DR}_t)^{(Y-BY+0.5)}} \\ & + \sum_y \frac{\text{Cost}_{I,y} \cdot (\sum_d \text{RepW}_d \cdot \sum_{i,h} \text{imp}_{y,i,d,h}) - \text{Price}_{E,y} \cdot (\sum_d \text{RepW}_d \cdot \sum_{i,h} \text{exp}_{y,i,d,h})}{(1 + \text{DR}_t)^{(Y-BY+0.5)}} \end{aligned} \quad (8)$$

3.2. Constraints

$$\text{cap}_{a,y,t} = \text{Cap}_{E,y,t} \forall_{y,t \in T^{\text{Ex}}} \quad (9)$$

$$\text{cap}_{a,y,t} = \sum_{y=\text{Lifetime}+1}^Y \text{Cap}_t \cdot \text{inv}_{y,t} \forall_{y,t \in T^{\text{New}}} \quad (10)$$

$$\sum_y \text{inv}_{y,t} \leq 1 \forall_{t \in T^{\text{New}}} \quad (11)$$

$$\sum_D \text{RepW}_d \cdot \sum_h \sum_{t \in T^{\text{RE}}} \text{pg}_{y,t,d,h} \geq \text{MSRE}_y \cdot \sum_d \text{RepW}_d \cdot \sum_h \text{PD}_{y,d,h} \forall_y \quad (12)$$

$$\begin{aligned} \sum_{T^D} \text{cap}_{a,y,t} \cdot (1 - \text{FOR}_t) + \sum_{T^{\text{ND}}} \text{cap}_{a,y,t} \cdot \text{mean}(\text{Cap}_{\text{Fct},t,d,h}) \\ \geq (1 + \text{RMR}) \cdot \max_{d,h} (\text{PD}_{y,d,h}) \forall_y \end{aligned} \quad (13)$$

$$\text{cmt}_{y,t,d,h} \leq \text{cap}_{a,y,t} \forall_{y,t \in T^D,d,h} \quad (14)$$

Constraints (9)-(13) concern investment decisions. Accounting for the lifetime of each unit, the available capacity $\text{cap}_{a,y,t}$ for existing and new units in each year is respectively formulated by equations (9) and (10). Constraint (11) ensures that the construction of a new unit is at most occurred once during the planning horizon. Constraint (12) imposes the minimum share of renewable energy in each year by weighting daily renewable generation and power demand. The reserve margin constraint over the planning horizon is forced by (13). Constraint (14) depicts the relation between commitment and investment decisions for dispatchable units. This relationship for non-dispatchable units is shown in constraint (16).

$$\sum_{t \in T^D} \text{pg}_{y,t,d,h} + \sum_{t \in T^{\text{ND}}} \text{pg}_{y,t,d,h} + \sum_i \text{imp}_{y,i,d,h} = \text{PD}_{y,d,h} + \sum_i \text{exp}_{y,i,d,h} \forall_{y,d,h} \quad (15)$$

$$\text{pg}_{y,t,d,h} \leq \text{Cap}_{\text{Fct},d,h} \cdot \text{cap}_{a,y,t} \forall_{y,t \in T^{\text{ND}},d,h} \quad (16)$$

$$\text{pg}_{y,t,d,h} - \text{res}_{\text{dry},t,d,h} \geq \text{Cap}_t \cdot \text{MSG}_t \cdot \text{cmt}_{y,t,d,h} \forall_{y,t \in T^D,d,h} \quad (17)$$

$$\text{pg}_{y,t,d,h} + \text{res}_{\text{ur},y,t,d,h} + \text{res}_{\text{us},y,t,d,h} \leq \text{Cap}_t \cdot \text{cmt}_{y,t,d,h} \forall_{y,t \in T^D,d,h} \quad (18)$$

$$\text{stu}_{y,t,d,h} - \text{std}_{y,t,d,h} = \text{cmt}_{y,t,d,h} - \text{cmt}_{y,t,d,h-1} \forall_{y,t \in T^D,d,h} \quad (19)$$

$$\text{stu}_{y,t,d,h} + \text{std}_{y,t,d,h} \leq 1 \forall_{y,t \in T^D,d,h} \quad (20)$$

$$\sum_h \text{MDT}_{t-1} \text{cmt}_{y,t,d,h} \leq (1 - \text{std}_{y,t,d,h}) \cdot \text{MDT}_t \forall_{y,t \in T^D,d,h} \quad (21)$$

$$\sum_h \text{MUT}_{t-1} \text{cmt}_{y,t,d,h} \geq \text{stu}_{y,t,d,h} \cdot \text{MUT}_t \forall_{y,t \in T^D,d,h} \quad (22)$$

$$\text{pg}_{y,t,d,h+1} - \text{pg}_{y,t,d,h} \leq \text{RUL}_t \forall_{y,t \in T^D,d,h} \quad (23)$$

$$\text{pg}_{y,t,d,h} - \text{pg}_{y,t,d,h+1} \leq \text{RDL}_t \forall_{y,t \in T^D,d,h} \quad (24)$$

$$\sum_{t \in T^{\text{DR}}} \text{res}_{\text{ur},y,t,d,h} \geq \text{Res}_{\text{URF}} \cdot \text{PD}_{y,d,h} \forall_{y,d,h} \quad (25)$$

$$\sum_{t \in T^{\text{DR}}} \text{res}_{\text{dr},y,t,d,h} \geq \text{Res}_{\text{DRF}} \cdot \text{PD}_{y,d,h} \forall_{y,d,h} \quad (26)$$

$$\sum_{t \in T^{\text{DR}}} \text{res}_{\text{us},y,t,d,h} \geq \text{Res}_{\text{USF}} \cdot \text{PD}_{y,d,h} \forall_{y,d,h} \quad (27)$$

$$\text{imp}_{y,i,d,h} \leq \text{IC}_{\text{Imp}_i} \forall_{y,i,d,h} \quad (28)$$

$$\text{exp}_{y,i,d,h} \leq \text{IC}_{\text{Exp}_i} \forall_{y,i,d,h} \quad (29)$$

Constraints (15)-(29) relate to system operation. Considering power import and export, load balance at each hour of representative days is imposed by equation (15). Constraint (16) determines power output from non-dispatchable units through capacity factor. To permit renewable power curtailment, the equality constraint is changed to the inequality in (16). Power output from dispatchable units is limited by constraints (17) and (18). Start-up and shutdown decisions in each hour are formulated by constraints (19) and (20). Constraints (21) and (22) respectively impose the minimum down and up time limitations. Ramp rate constraints in both directions are forced by (23) and (24). The formulations for upward regulating, downward regulating and upward contingency spinning reserves are respectively depicted by constraints (25)-(27). The amount of contribution of each dispatchable unit that can provide reserve $t \in T^{\text{DR}}$ is calculated by constraints (17) and (18). Constraints (28) and (29) specify that the amount of power import and export at each hour, computed by equation (15), must be in accordance with the capacity of interconnectors.

4. Test systems

Two power systems were examined to show the effectiveness of the proposed methods. One of them was NREL-118 [42], in which VRE technologies comprise a considerable number of generators. Hourly time series of the power system, including load, wind power and solar PV power, have also been provided for one year, which was the rationale for employing NREL-118 in this study. The Belgian electricity system, as an instance of a real power system, was considered for the second case. A way to take advantage of several years of net load data was introduced for the Belgian case.

The outcomes of nine approaches to selecting representative days were used in the planning model, and the obtained results from GEP were compared. These approaches include the conventional COPT, CWard and CPAM methods, in which just load and VRE generation time series took part. These are the conventional methods that are used in the literature. The other approaches are the integration of the Opt, Ward and PAM with the proposed way of incorporating the extreme days of the net load, which are called SOM&Opt, SOM&Ward and SOM&PAM in the rest of the paper and have been introduced in Section 2.1. To investigate the effect of proper weight determination of the extreme days by using the SOM clustering, the Opt, Ward and PAM are also integrated with two specific days during which the minimum and

maximum of the net load have occurred. These approaches are respectively named H&Opt, H&Ward and H&PAM where H stands for heuristic. The weight of each of the extreme days is one in these approaches because the number of their similar days is unknown. Besides, the integration process is the same as the procedure described for the SOM clustering in Section 2.1.3. As a result, two approaches to assigning weights to the extreme days of the net load are pursued in this study, including the SOM clustering that is described in Section 2.1.2 and the heuristic approach described here.

It is also noteworthy that we are interested in the improvement that using the extreme days would bring about in selecting representative days. Thus, the outcomes of those planning models were compared with each other for which the same approach had been used to select representative days from load and VRE generation, such as the COpt, SOM&Opt and H&Opt. A summary of the methods for selecting representative days is provided in Table 1.

In addition, the number of representative days should be identical since deploying the higher number of days in the planning model would cause more accurate results. Specifically, the suitable number of representative days was individually determined for the COpt, CWARD and CPAM methods by the rigorous and acceptable manners, explained in Section 2.2. The number of days for the other methods employing the SOM or heuristic approaches was then set the same as that for the COpt, CWARD and CPAM. For instance, if eight days were found suitable for the CPAM, the SOM&PAM and H&PAM were consequently adjusted to selecting eight representative days while two days were respectively chosen by the SOM or heuristic approach from the net load. The other six days in this example should be selected by the PAM clustering algorithm from the load and VRE data. In the case of the SOM&Opt and H&Opt, if, for example, ten days were identified proper for the COpt, the weights of the chosen ten days were optimized in a way that only eight of them had a value, so the other two representative days were picked from the net load.

The GEP model described in Section 3 was employed for both test systems. All the optimization problems were written in the GAMS and solved by the CPLEX solver. The relative optimality gap of 0.01 was set for MILP problems. All the clustering methods and the proposed process explained in Section 2.1 were written in the R programming language [43]. The Ward, PAM and SOM methods were respectively performed by the dtwclust [44], cluster [45], and kohonen [46], packages. The dtwclust package was also used for calculating the DTW distance measure. All runs are performed on a system with an Intel Core i5-4590S CPU @ 3.00 GHz and 8.00 GB of RAM.

4.1. NREL-118 test system

4.1.1. Data and assumptions

The planning model is executed for one year for the NREL-118 test system, and the effect of the extreme days on both investment and operational decisions are investigated. The time series of wind, solar PV and load, with a maximum of around 5300 MW, from the third region of the NREL-118 [42], was used. These hourly time series have been provided for one year, and about 22 percent of demand is supplied by VRE technologies in this region.

The model is run in a greenfield mode, with no existing units at the beginning of the planning horizon. The set of new units was collected from an EIA document [47]. The up-to-date, economic information on the units was gathered from the EIA document for 2019 [48]. The operational characteristics were obtained from Gonzalez-Salazar et al.'s work [49], and Schröder et al.'s work [50], or from the original equipment manufacturers' documents. The operational and economic features of the new units are respectively exhibited in Tables 2 and 3.

Solar PV and wind technologies were the only non-dispatchable units, and all the dispatchable power plants could take part in providing the operating reserves. Interconnection lines with capacity of 600 MW were used for power import and export. To fully employ the installed

capacity in the region, the cost of power import and price for power export were respectively decided to be equal to the highest and lowest figures for variable costs of power generation among dispatchable units. The price of natural gas (NG) and coal were set 5.4 and 1.8 $\frac{\$}{\text{MMBtu}}$ respectively. The upward and downward regulating reserve and contingency spinning reserve were equal to 0.01 and 0.03 of load respectively [42]. The minimum penetration of renewable resources was set 0.2 of the annual electricity consumption.

4.1.2. Results and discussion

The effect of different methods for selecting representative days (see Table 1) on the optimal capacity mix and operational decisions is analyzed in this section. The planning results of various methods were compared to those of a full year. The units shown in Table 2, except for biomass and advanced CC, were utilized as the new units.

The number of representative days was determined by the rigorous and acceptable methods, described in Section 2.2. For the CWARD and CPAM, the number of clusters varied from 2 to 20, while for the COpt the even numbers from 2 to 10 were examined due to the computational complexity of the Opt method. The number of bins and randomly selected subsets of days were respectively 50 and 20,000 for the COpt.

Fig. 2 illustrates how the rigorous and acceptable manners work in order to get the number of representative days, which was carried out for the CPAM. The yellow dots in Fig. 2, the minimum of each metric, display the options for the number of days according to the rigorous manner. The greatest day index among these options was regarded as the suitable number of representative days. The red dots are the first days below the grey line that is the sum of the minimum and standard deviation of each metric. These are the options for the number of days according to the acceptable manner. The greatest day index among the options was deemed to be the accepted number of representative days. In the remainder of the paper, the symbols of the methods, described in Table 1, are followed by an underscore with a number that indicates the number of representative days. It is obvious that the smaller number of days pertains to the acceptable manner, and the higher ones are found by the rigorous manner.

As mentioned earlier, the methods with the same number of days besides the same type of approach among the Opt, Ward and PAM should be compared with each other. Fig. 3 displays the optimal generation technology mix obtained from various methods. It is obvious that the conventional COpt, CWARD and CPAM approaches underestimated the required capacity. More specifically, VRE capacity for different methods was similar to that for the full year, whereas a major discrepancy in thermal capacity can be seen when the extreme days of the net load are not included. Indeed, the full year and approaches deploying the extreme days determined around 4300 MW of dispatchable capacity, but COpt, CWARD and CAPM identified roughly 3500 MW. This deviation is also reflected in the estimation of the investment costs. While COpt, CWARD and CPAM underestimated the

Table 1
A summary of approaches used for representative days selection.

Method	Time series used	Explanation
COpt	Load	All of the representative days are selected by using load and VRE time series.
CWARD	VRE	
CPAM	generation	
SOM&Opt	Net load	Extreme days of the net load are found by the SOM. The other representative days are then selected by the specified approach through the use of load and VRE time series.
SOM&Ward	Load	
SOM&PAM	VRE	
H&Opt	Net load	Days during which the minimum and maximum of net load have occurred are considered as the extreme days. The other representative days are then selected by the specified approach through the use of load and VRE time series.
H&Ward	Load	
H&PAM	VRE	
	generation	

Table 2
Operational characteristics of the new units for the NREL-118 case and the Belgian power system.

Type of unit and fuel	Cap (MW)	MSG (%)	MDT (hour)	MUT (hour)	RDL/RUL (MW/hour)	Heat rate (Btu/kWh)	CO2 Emission factor (lb/MWh)
Coal	650	0.37	8	11	318	9750	1404
CC NG (Conventional)	702	0.22	4	8	1200	6600	772.2
CC NG (Advanced)	1100	0.22	4	8	1200	6300	737.1
CT NG (Conventional)	100	0.25	1	1	720	9840	1151.3
CT NG (Advanced)	237	0.3	2	2	1200	9800	1146.6
Bio	50	0.4	6	8	60	13,500	2632.5
Solar PV	150	–	–	–	–	–	0
Wind	100	–	–	–	–	–	0

Table 3
Economic characteristics of the new units for the NREL-118 case and the Belgian power system.

Type of units and fuel	Capital costs (\$/kW) [48],	Fixed O&M (\$/MW) [48],	Variable O&M (\$/MWh) [48],	Start-up costs *(\$/Cap) [51],
Coal	5169	72,120	7.31	104
CC NG (Conventional)	999	11,330	3.61	79
CC NG (Advanced)	794	10,300	2.06	79
CT NG (Conventional)	1126	18,030	3.61	32
CT NG (Advanced)	691	7010	11.02	103
Bio	3900	114,390	5.7	5.3
Solar PV	1783	22,460	0	0
Wind	1624	48,420	0	0

* Start-up costs have been calculated in 2011 US dollars.

needed costs for capacity expansion with around 22–35 percent, the other approaches deploying the extreme days accurately estimated the capital costs with less than 1.5 percent error. Better estimation of system costs by using the extreme values of the net load has also been observed by Scott et al. [29]. These findings illustrate that the extreme days, especially days with higher demand and lower VRE generation, are crucial to capacity planning because more dispatchable capacity is needed to compensate for lower generation from VRE in these days. According to Fig. 3, finding and assigning weights to the extreme days either by the SOM or heuristically did not make a big difference because the capacity mix and investment costs obtained from these approaches is almost the same. In addition, the contrast between using the rigorous or acceptable manner to select the number of representative days was minimal, and both of them performed well when the extreme days are included in representative days.

In Table 4, the annual variable operating costs of the system, which is equal to the sum of variable operating costs of the units and the costs of net import, are compared between different approaches to selecting representative days. This criterion is picked because it is not directly affected by the installed capacity, and it only depends on the operating decisions. It is evident that all of the COpt, CWard and CPAM methods extremely overestimated the variable operating costs. In this case, lack of coal power plants in the generation fleet determined by the mentioned methods caused this huge deviation. As a result, power generation switched to the CC and combustion turbine (CT) units that employ NG, which is far more expensive than coal in this case. By looking at the figures for the methods employing the extreme days of the net load, it is understood that using the SOM clustering to find and weight the extreme days led to a much better approximation of the operating costs for all the cases. The relative improvement in the estimation of the operating expenses owing to assigning a proper weight to the extreme days is different between the methods and number of days, but the average magnitude of the relative improvement is about 45 percent. It means that by giving more accurate weights to the extreme days by using the SOM clustering, instead of setting the weights equal to 1 for each extreme day, 45 percent enhancement in the estimation of

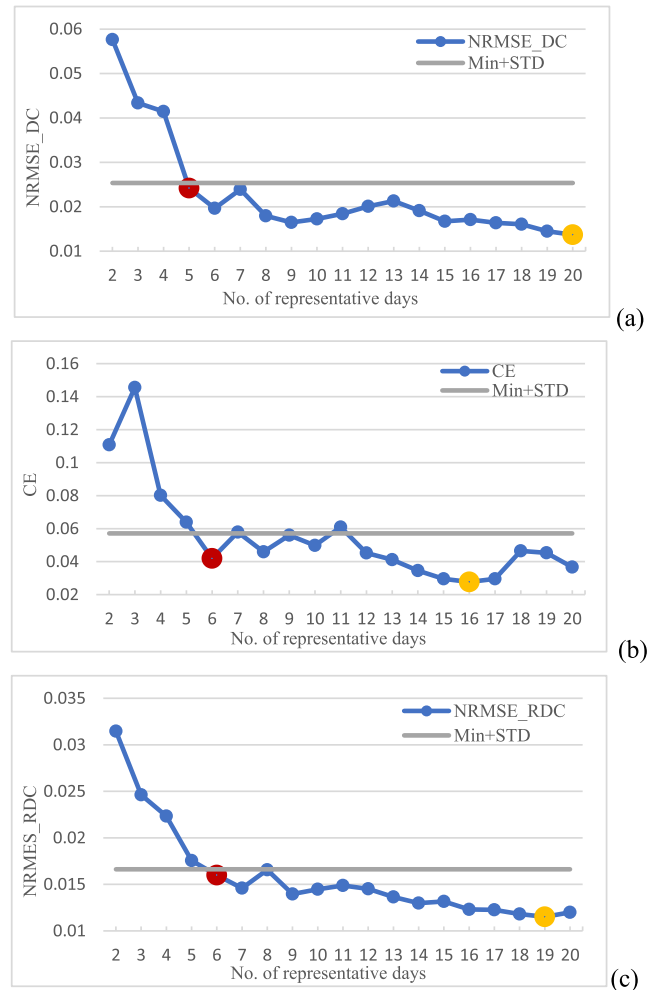


Fig. 2. Approximation metrics (a) $NRMSE_{av}^{DC}$, (b) CE_{av} and (c) $NRMSE_{av}^{RDC}$ calculated for various number of representative days for the CPAM, and two suggested ways to choose the number of representative days.

operating costs would be expected on average. It is also observed that both suggested ways to choose the number of representative days appear to work well when the extreme days are included. By comparing the results of the COpt, CWard and CPAM methods with different number of representative days, it is also inferred that increasing the number of representative days does not compensate for not using the extreme days because the required capacity and operating costs were underestimated by these methods even with the higher number of representative days.

Considering the required dispatchable capacity and operating costs, we concluded that COpt, CWard and CPAM are not good options for picking representative days. Although using the SOM clustering to give weights to the extreme days performed much better in the calculation

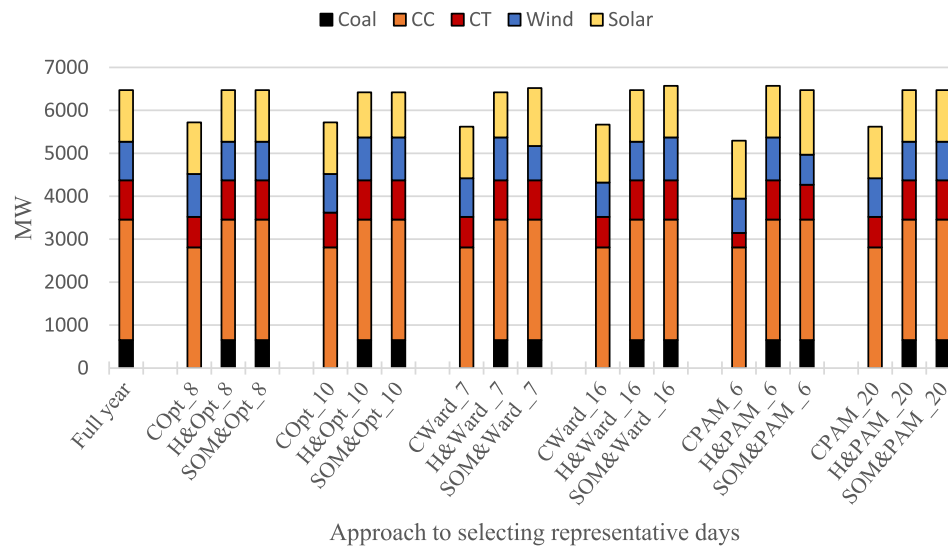


Fig. 3. The optimal generation technology mix resulted from various methods for selecting representative days.

Table 4

The system operating costs for different methods of selecting representative days.

Method	Annual variable operating costs(Million \$)	Error (%)
Full year	698.2	–
COpt_8	793.2	13.60
H&Opt_8	695.2	0.42
SOM&Opt_8	696.5	0.24
COpt_10	792.0	13.43
H&Opt_10	695.6	0.37
SOM&Opt_10	697.4	0.12
CWard_7	801.4	14.77
H&Ward_7	704.6	0.91
SOM&Ward_7	693.5	0.67
CWard_16	785.5	12.50
H&Ward_16	687.1	1.59
SOM&Ward_16	700.1	0.34
CPAM_6	795.4	13.92
H&PAM_6	703.4	0.74
SOM&PAM_6	702.8	0.66
CPAM_20	794.9	13.85
H&PAM_20	694.8	0.48
SOM&PAM_20	696.3	0.28

of the operating costs, further investigation should be conducted to reach a conclusion. Thus, three other criteria are analyzed to compare the performance of the approaches to weighting the extreme days. All of these criteria pertain to the system operating variables, including total CO₂ emissions besides the annual generation from VRE and CT units. The first two indicators are picked owing to the importance of VRE generation and CO₂ emissions in power systems with high penetration of renewable resources. The latter is a great option for comparison because flexible CT units have a vital role in days with higher and lower values of the net load for load balancing.

The magnitude of errors in calculating the annual generation from CT units, obtained from the approaches incorporating the extreme days of the net load, is displayed in Fig. 4. The SOM approaches to weighting the extreme days calculated the CT generation much more accurately, with errors less than 16.5 percent. In contrast, all the other approaches, in which the extreme days are selected manually with the weight of 1, extremely underestimated the CT generation with errors ranging from 28 to 52.5 percent. For calculating the annual generation, the power generation in representative days is multiplied by their corresponding weights. Accordingly, the CT generation was hugely underestimated for the heuristic (H) methods because the weight of 1 is multiplied by CT generation in the extreme days. Similarly, the better estimation of the

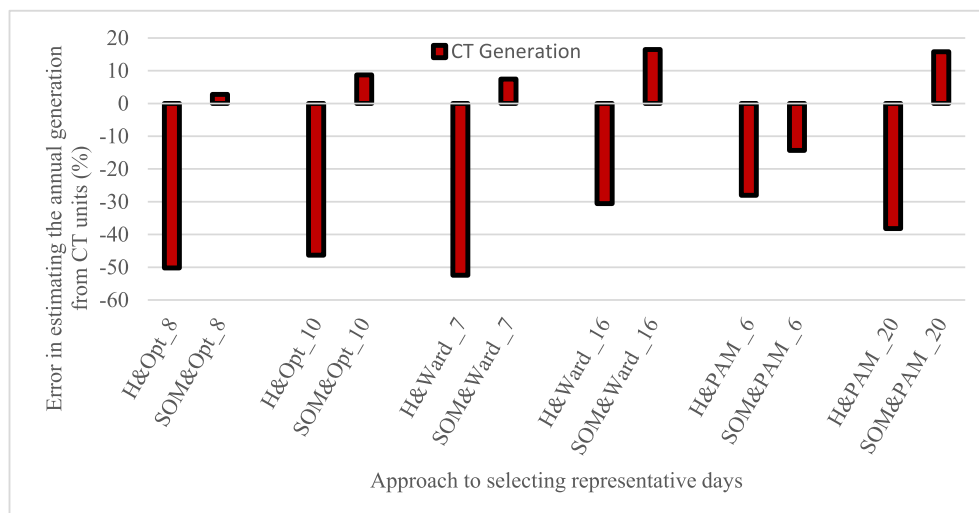


Fig. 4. Errors in estimating the annual generation from CT units obtained from approaches incorporating the extreme days.

CT generation by the SOM approaches would be a result of the weights that represent the actual number of the extreme days in the historical data.

The errors in estimating the annual generation from VRE units and CO₂ emissions are exhibited in Fig. 5 for approaches incorporating the extreme days. The magnitude of the errors is less than 1.7 percent, which demonstrates the effectiveness of both approaches to approximating both operating variables. However, using the SOM to allocate weights to the extreme days resulted in better estimations of the annual CO₂ emissions in all the approaches, except for the SOM&PAM₆. A more accurate approximation of the yearly VRE generation was also obtained for all the cases, except for the SOM&Ward₇, by using the SOM approaches. Generally, when both indicators are considered, the SOM&Opt₈, SOM&Opt₁₀, SOM&Ward₇ and SOM&PAM₂₀ are preferred to their heuristic (H) counterpart. For the rest of the approaches, including the Wards with 7 days and PAMs with 6 days, a conclusion cannot be drawn by considering only CO₂ emissions and the VRE generation since the direction of changes in these two indicators is the opposite when different weighting approaches are employed. In terms of the number of representative days, a significant difference cannot be seen in the operating indicators when either the acceptable or rigorous manner were implemented, and it can be said that both of them performed satisfactorily.

The average errors in calculating the operating variables are provided in Table 5 for the SOM and heuristic (H) methods of weighting the extreme days. For the CT generation, the magnitude of the average errors is much higher for both methods compared to the other two indicators. However, using the SOM to weight the extreme days immensely improved the estimation of the annual CT generation, with 73 percent of enhancement in the average errors. The average errors in estimating the annual CO₂ emissions and VRE generation were also much lower when the SOM was used to assign weights to the extreme days. The relative errors in approximating the yearly CO₂ emissions and VRE generation respectively improved by 29.7 and 28 percent when the SOM method was used. Generally, it is inferred that the underlying reason for the average 45 percent improvement in calculating operating costs by the SOM is that this approach to weighting the extreme days could approximate the operating variables, such as generation from different units, more accurately. It is also noteworthy that the mentioned improvements by the SOM approach are observed for single-year planning, and it would be expected that these improvements become more significant in planning for multiple years.

Table 5

The average errors in calculating the annual operating variables for the SOM and heuristic methods of weighting the extreme days.

Approach to weighting the extreme days	The average error in estimating the annual amount of (%)		
	CT generation	CO ₂ emissions	VRE Generation
Heuristic (H)	40.96	0.91	0.90
SOM	10.92	0.65	0.63
The percent of the relative improvement by using the SOM instead of H	73.3	29.7	28.0

4.2. The Belgian power system

4.2.1. Data and assumptions

Another power system is analyzed in this work for two reasons. Firstly, the net load time series depends on VRE capacity, so finding the extreme days of the net load is complicated when VRE capacity changes over the historical period, which is common in a real case. The application of the proposed method for selecting representative days, as a result, was shown for several years of historical data in this section. Secondly, the effect of using the extreme days of the net load, with different weighting approaches, on multi-year planning was investigated for a real case with high penetration of VRE.

The hourly load, solar PV and wind time series from 2016 to 2018 for the Belgian power system were gathered from the Belgian transmission system operator Elia [52]. It is unreasonable to compare days with different levels of VRE capacity in order to obtain the extreme days out of the net load because the net load is equal to load minus VRE capacity times VRE capacity factor. Presenting the relation between load and capacity factor of VRE resources is a unique feature of the net load. Thus, we suggest that VRE capacity factor should be multiplied by an identical figure for VRE capacity over the whole of the historical period so that the net load features are preserved, and its levels become comparable. Hence, it was assumed that capacity of VRE resources over the three-year period had been equal to the VRE capacity at the end of the period. Then, the net load at each hour was computed by subtracting the multiplication of capacity factor of each VRE resource and the corresponding capacity at the end of 2018 from load. The capacity at the end of the historical period was considered because it forms the existing capacity of the planning problem, and new VRE capacity would be added to these figures. After the net load was adjusted, the

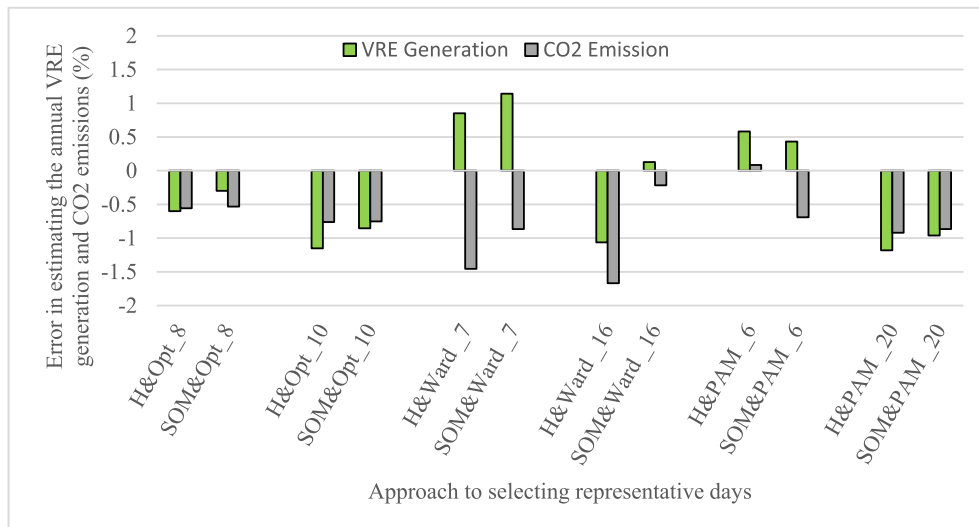


Fig. 5. Errors in estimating the annual generation from VRE units and CO₂ emissions obtained from approaches incorporating the extreme days.

approaches to selecting representative days can be implemented without any changes. The obtained weights of representative days must be divided by the number of years so that their sum equals the number of days in a year.

Seven years from 2019 to 2025 were examined by the planning model. The existing units were taken from Elia documents [52]. Some modifications were made to the generation fleet such as replacing pumped storage and combined heat and power units with CT units. The operational and economic features of the new units are respectively presented in Tables 2 and 3. All the new units except for coal power plants are options for capacity expansion. The operating reserve and price for power export were identical to those in the NREL test system, while the cost of power import was decided to be equal to the third quartile of figures for variable costs of power generation among dispatchable units. The only non-dispatchable units were solar PV and wind power plants, and all the dispatchable units could participate in providing operating reserves except for nuclear power plants. The price of NG, nuclear and biomass fuels were set 7.5, 2.4 and $10 \frac{\$}{\text{MMBtu}}$ respectively. Annual growth of load was considered 0.6%. The figures for the rest of the parameters are described in Table 6.

4.2.2. Results and discussion

In contrast to the previous test system, we focus on multi-year planning in this section to analyze the differences in capacity planning that could be resulted from various approaches to selecting representative days. In the following paragraphs, each of the Opt, Ward and PAM approaches is studied separately. Additionally, two methods of selecting the number of representative days performed almost the same according to the previous test system, so only the acceptable manner was employed here.

The accumulated, new capacity added to the generation fleet is shown in Fig. 6. for Opt approaches to selecting representative days. Just like the results of the NREL case, the COpt led to the underestimation of dispatchable capacity, which is more noticeable in the year 2025 by determining 700 MW less of CC capacity. As a result, the discounted investment costs of the COpt was 5.6 percent lower compared to the SOM&Opt because more dispatchable capacity is required to satisfy the net load in the extreme days. In terms of weighting the extreme days, the required capacity at the end of planning was the same for both the SOM&Opt and H&Opt. However, it was decided to install more dispatchable capacity in 2021 by the SOM&Opt. Indeed, it was required to install CC plants sooner when the SOM approach assigns the weights of the extreme days. The need to install dispatchable capacity sooner for the SOM&Opt is also reflected in the total capital costs discounted to the base year (see Table 7) because this figure for dispatchable units was underestimated by 3.3 percent when the H&Opt was applied instead of the SOM&Opt. These findings highlight that weight 1 for the extreme days of the net load is not enough to force the planning model to determine dispatchable capacity whenever needed in the planning horizon.

In Fig. 7, the accumulation of the capacity expansion is displayed for methods of selecting representative days using Ward's algorithm to select the normal days. Both of the findings for the Opt method, mentioned above, are discerned in Fig. 7. The required thermal capacity was much lower throughout the planning horizon when the net was not incorporated in selecting representative days, leading to an underestimation of 15.7% for discounted capital investment in dispatchable units. Indeed, the extreme days of the net load, especially days with higher net load levels, are a decisive factor in expansion planning because more dispatchable capacity is needed for these days to satisfy the proportion of load that is not met by non-dispatchable units. Further, the SOM&Ward determined the required dispatchable capacity earlier in the planning horizon compared to the H&Ward. It is the reason why the discounted costs of investing in dispatchable units for H&Ward is 0.6% lower than those for the SOM&Ward in Table 7. These results

reveal the importance of weighting the extreme days in power system planning and confirm that weight 1 for the extreme days of the net load is not sufficient. The SOM gives weights to extreme days according to their importance and the number of similar days; as a result, more dispatchable capacity and a more reliable system were determined throughout the planning horizon by the SOM approach.

The accumulation of the needed capacity for different PAM methods for selecting representative days is exhibited in Fig. 8. The results concerning the dispatchable units were similar to those for the Opt and Ward methods. As a result, the costs of investing in dispatchable units for the CPAM and H&PAM were respectively 14% and 2.7% lower compared to the SOM&PAM. Nonetheless, a significant difference in the added VRE capacity can be seen in Fig. 8. Throughout the planning horizon, much more VRE capacity was determined by the CPAM and H&PAM, which is also reflected in investment costs for VRE units in Table 7. For further evaluation, VRE generation over the planning horizon was calculated for all the approaches, and it was noticed that VRE generation for the CPAM and H&PAM was roughly 0.6% more than that for the SOM&PAM. Indeed, more VRE capacity for the CPAM and H&PAM brought about only a minor extra generation from VRE plants. It is also confirmed by the same constraint for renewable energy penetration and similarity in the installed biomass plants and generation from them for all the methods. We use the findings of the NREL test system to explain what the reason for lower installed capacity of VRE for the SOM&PAM could be. In the NREL case, it was observed that around 28 percent improvement in estimating the annual VRE generation could be achieved by using the SOM approach to weighting the extreme days of the net load, instead of the heuristic (H) one. Generation from VRE units in each extreme day is multiplied by a weight that is not reflecting the number of similar days to an extreme day when the heuristic (H) approach is used. Thus, it is more reliable to use the SOM approach to weighting the extreme days because the weights reflect the importance of each of the extreme days in the historical data. Generally, since generation from VRE plants in the extreme days of the net load is different from the normal days, leading to higher and lower values of the net load, accurate weighting of the extreme days by the SOM would cause a more reliable estimation of the required VRE capacity. Regarding the difference in VRE capacity between the CPAM and SOM&PAM, it is true that the CPAM does not take account of the extreme days and VRE generation in these days at all, so the estimation of the needed capacity of the VRE units as well as dispatchable plants would be inaccurate. These results about the needed VRE installation were also observed for the Opt methods, with a lower magnitude, and the same explanation discussed here is also valid for them.

4.3. The computational cost

In this subsection, the computational cost of implementing different approaches to selecting representative days and the mathematical model is provided. The elapsed time for the execution of the methods for choosing representative days is displayed in Table 8 for the NREL-118 test system with 365 days. The SOM approaches to extracting the

Table 6
Parameters of the planning model for the Belgian case.

Parameter	Value
IC_Imp (MW) [52],	1800 (France to Belgium) – 950 (The Netherlands to Belgium)
IC_Exp (MW) [52],	600 (Belgium to France) – 950 (Belgium to The Netherlands)
MSRE (p.u.)	19% in 2019 increasing to 28.5% in 2025 (1.6% annual growth)
DR & DRI (p.u.)	0.07
FOR (p.u.)	CC (0.04) - CT (0.05) – NU (0.04) – Bio (0.09)
RMR (p.u.)	0.05

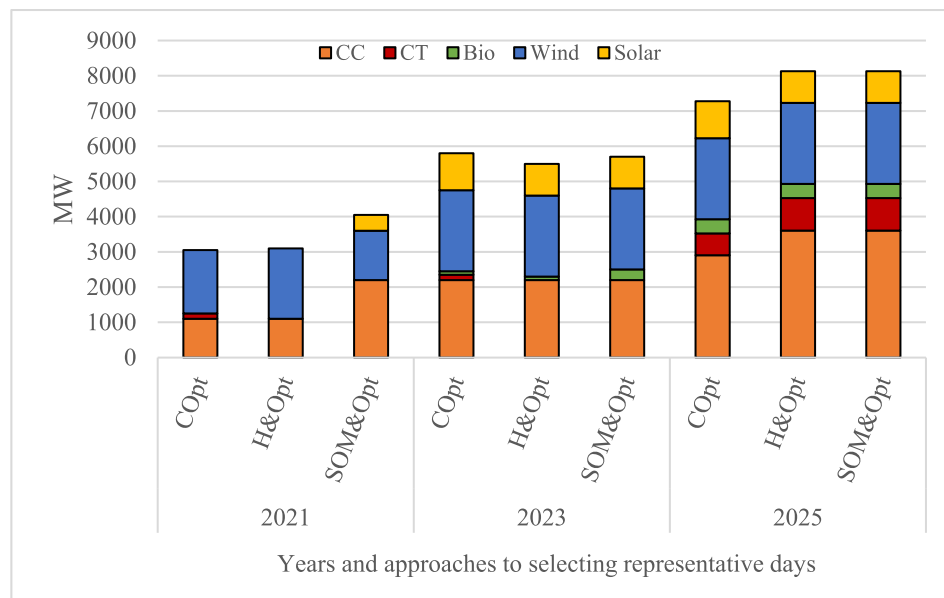


Fig. 6. The accumulation of the new, installed capacity in 3 years of the planning horizon for Opt approaches to selecting representative days, with 12 selected days.

Table 7

Total discounted investment costs of various approaches to selecting representative days.

Method	Discounted capital investment in (Billion \$)			Difference between the corresponding approach and the SOM one		
	Dispatchable units	VRE units	All the units	Dispatchable units	VRE units	All the units
COpt	3.52	8.13	11.64	−19.1%	1.7%	−5.6%
H&Opt	4.21	8.10	12.30	−3.3%	1.3%	−0.3%
SOM&Opt	4.35	7.99	12.34	–	–	–
CWard	3.57	9.16	12.73	−15.7%	−5.3%	−8.5%
H&Ward	4.21	9.46	13.67	−0.6%	−2.3%	−1.8%
SOM&Ward	4.24	9.68	13.91	–	–	–
CPAM	3.62	9.51	13.13	−14.0%	8.8%	1.4%
H&PAM	4.09	10.8	14.87	−2.7%	23.3%	14.8%
SOM&PAM	4.21	8.74	12.95	–	–	–

extreme days brought about an extra computational burden, which is equal to 13 min in this case. However, the advantages gained by the SOM approaches in power system planning, such as determining a more reliable system, overshadow their higher computational costs, especially by considering the fact these methods are employed for only one time to select representative days.

The time required to find representative days in the Belgian case was around three times as high as that in the NREL-118 case for each of the methods. The reason is that representative days were selected from three years of historical data for the Belgian power system.

In terms of the GEP model, it took about 70 min to solve the full-year planning for the NREL-118 test system, but GEP models with representative days were executed in less than a minute for this case. For the Belgian power system, the average elapsed time for solving GEP was respectively around 23, 46, and 12 min for each of the approaches employing the Opt, Ward, and PAM methods. The differences in the execution time were due to the number of representative days used in GEP.

5. Conclusion

In this study, a new approach was proposed to select representative

days by taking advantage of the net load besides load and variable renewable energy (VRE) generation curves. The idea for this approach is that extreme days of the net load, i.e. days with higher and lower levels of the net load, must be included in representative days since more capacity is needed to satisfy load during the extreme days. Self-organizing map (SOM) clustering was employed to find the extreme days of the net load automatically. To select the rest of representative days by using load and VRE curves, other methods of selecting representative days, such as Ward's and PAM algorithms, can be integrated with our proposal. One of the advantages of the proposed approach is that the number of similar days in the historical data determines the weights of the extreme days. However, another variation of choosing the extreme days is suggested, in which days with the peak and trough of the net load are heuristically regarded as the extreme days, and each of them receive weight 1. Accordingly, three various types of approaches to selecting representative days were compared for planning of two different test systems. These types include the conventional methods by using only load and VRE time series, the combination of the SOM approach for the extreme days and the conventional ones, and the combination of the heuristic approach to choosing the extreme days and the conventional methods. As a result, it became possible to investigate the effect of both the extreme days and weighting of them on power system planning.

In addition, levels of the net load vary by the addition of VRE units over years, so the way to use multiple years of the net load for representative days selection is offered for the Belgian power system. Two methods were also introduced to select the proper number of representative days. These methods are based on how well a set of representative days approximates the load and VRE duration curves. Generally, the findings of this research can be categorized as follows:

- 1) The extreme days of the net load are not included in the representative days that are chosen by the conventional methods. By using these methods, the required dispatchable capacity and investment costs are severely underestimated in power system planning since the extreme days do not have any effect on system expansion.
- 2) The inclusion of the extreme days of the net load in representative days leads to a much more accurate estimation of capital cost, the optimal generation capacity mix, and the operating costs. However, using the SOM approach to weighting the extreme days improves

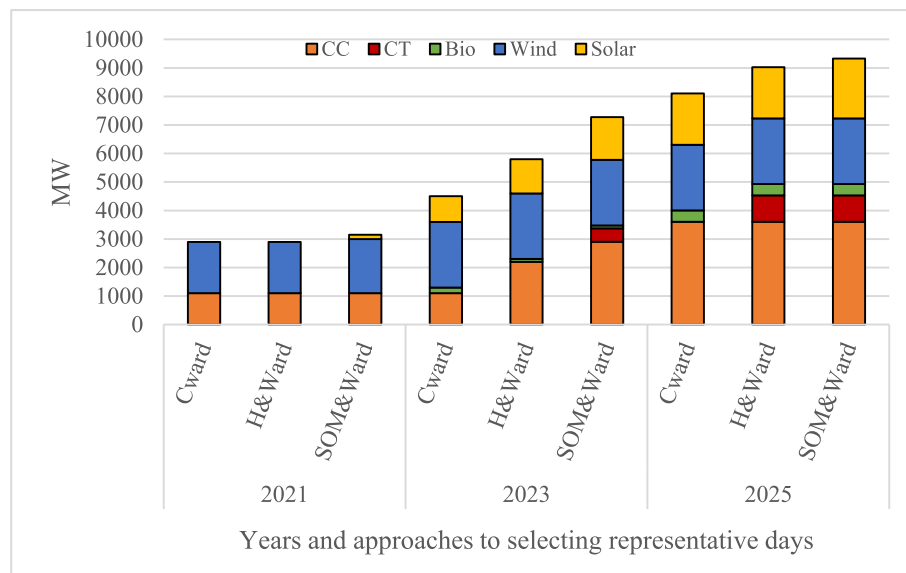


Fig. 7. The accumulation of the new, installed capacity in 3 years of the planning horizon for Ward approaches to selecting representative days, with 14 selected days.

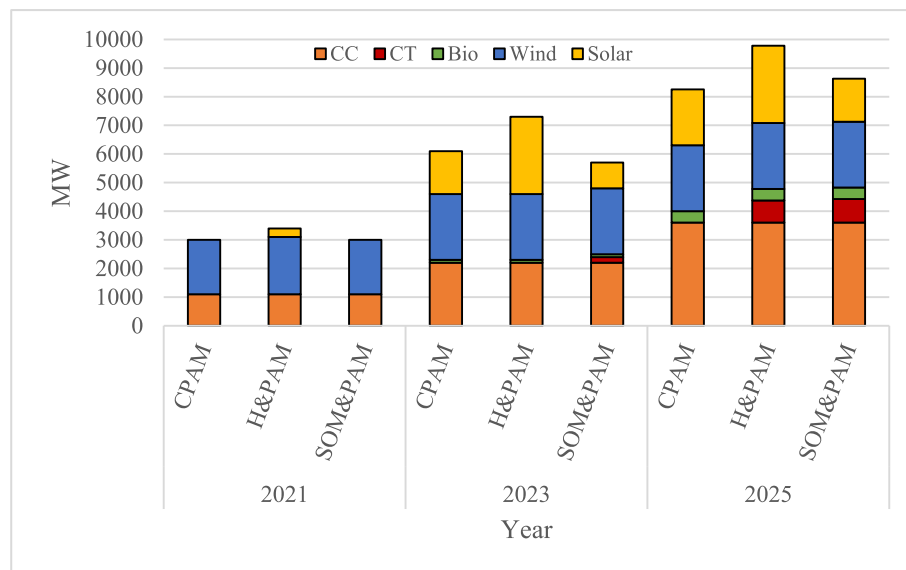


Fig. 8. The accumulation of the new, installed capacity in 3 years of the planning horizon for PAM approaches to selecting representative days, with 8 selected days.

Table 8

The elapsed time of the implementation of the methods for selecting representative days for the NREL-118 case.

Method	Elapsed time (minute)
COpt	122.20
H&Opt	122.20
SOM&Opt	135.23
CWard	0.01
H&Ward	0.01
SOM&Ward	13.03
CPAM	0.01
H&PAM	0.01
SOM&PAM	13.03

the calculation of the operating costs by around 45 percent compared to its heuristic counterpart.

- 3) Further evaluation showed that operating variables are estimated a lot better when the weights of the extreme days are determined by

the SOM approach. More specifically, the approximation of the annual generation from combustion turbine units, generation from VRE plants and CO₂ emissions respectively improved by about 73, 28 and 30 percent on average by employing the SOM approach instead of the heuristic one. The SOM approach, therefore, is strongly recommended for temporal representation when the impact of operational policies, such as curtailment reduction and demand response, on the planning decisions are considered being analyzed.

- 4) Although the optimal generation capacity mix obtained from the planning for 1 year is the same for both methods of weighting the extreme days, more dispatchable units and capital investment in these units are determined for multi-year planning by the SOM approach. These findings highlight that weight 1 for the extreme days is not sufficient enough to force the model to determine the required dispatchable capacity whenever needed. Thus, weighting the extreme days according to the number of similar days in the historical data by the SOM results in more reliable planning for power systems.

- 5) Two methods for selecting the number of representative days worked well because capital costs, the required capacity and operating costs were estimated precisely for both manners when the extreme days were included. It is also much more imperative for power system planning to employ the extreme days compared to increasing the number of representative days.

The role of the extreme days of the net load in selecting representative days would become more significant when the share of VRE continues to grow in power systems. Moreover, the proposed method can be used to select periods other than a day, such as representative weeks. It also has the potential for selecting representative periods in a multi-regional power system.

Observing the differences between the single-year and multi-year planning in this research, we strongly recommend other researchers to analyze their improvements in generation expansion planning (GEP) for multiple year planning. Nevertheless, it is difficult to benchmark the results since there are no standard criteria for comparison, but in single year planning, the results can be compared to the full year planning. The future works can also focus on defining and finding the other extreme days automatically and investigating their impacts on power system planning.

On the other hand, future works could also concern the use of the proposed method for selecting representative days to address parameter uncertainties by stochastic programming or to model the interaction between agents in a liberalized power market by multi-level programming. In addition, developing a tractable GEP model holding a high temporal, technical, and spatial resolution can be a good option for future studies to investigate the effect of nodal variations of load and VRE.

CRediT authorship contribution statement

Ali Yeganefar: Conceptualization, Methodology, Software, Writing - original draft, Formal analysis, Visualization, Validation, Resources. **Mohammad Reza Amin-Naseri:** Writing - review & editing, Validation, Supervision, Resources. **Mohammad Kazem Sheikh-El-Eslami:** Writing - review & editing, Resources, Conceptualization, Validation, Methodology.

References

- IRENA. Global energy transformation: A roadmap to 2050. Abu Dhabi: International Renewable Energy Agency; 2018.
- Dagoumas AS, Koltsaklis NE. Review of models for integrating renewable energy in the generation expansion planning. *Appl Energy* 2019;242:1573–87.
- Cochran J, Miller M, Zinaman O, Milligan M, Arent D, Palmintier B, et al. Flexibility in 21st century power systems. Golden, CO (United States): National Renewable Energy Lab. (NREL) 2014.
- Alizadeh MI, Moghaddam MP, Amjadi N, Siano P, Sheikh-El-Eslami MK. Flexibility in future power systems with high renewable penetration: A review. *Renew Sustain Energy Rev* 2016;57:1186–93.
- Oree V, Hassen SZS, Fleming PJ. A multi-objective framework for long-term generation expansion planning with variable renewables. *Appl Energy* 2019;253:113589.
- Palmintier BS, Webster MD. Impact of operational flexibility on electricity generation planning with renewable and carbon targets. *IEEE Trans Sustain Energy* 2015;7:672–84.
- Palmintier B, Webster M. Impact of unit commitment constraints on generation expansion planning with renewables. 2011 IEEE power energy Soc. Gen. Meet., IEEE; 2011. p. 1–7.
- Poncelet K, Delarue E, Six D, Duerinck J, D'haeseleer W. Impact of the level of temporal and operational detail in energy-system planning models. *Appl Energy* 2016;162:631–43.
- Koltsaklis NE, Dagoumas AS. State-of-the-art generation expansion planning: A review. *Appl Energy* 2018;230:563–89.
- Mallapragada DS, Papageorgiou DJ, Venkatesh A, Lara CL, Grossmann IE. Impact of model resolution on scenario outcomes for electricity sector system expansion. *Energy* 2018;163:1231–44.
- Collins S, Deane JP, Poncelet K, Panos E, Pietzcker RC, Delarue E, et al. Integrating short term variations of the power system into integrated energy system models: A methodological review. *Renew Sustain Energy Rev* 2017;76:839–56.
- Koltsaklis NE, Dagoumas AS. Incorporating unit commitment aspects to the European electricity markets algorithm: An optimization model for the joint clearing of energy and reserve markets. *Appl Energy* 2018;231:235–58.
- Koltsaklis NE, Nazos K. A stochastic MILP energy planning model incorporating power market dynamics. *Appl Energy* 2017;205:1364–83.
- Nahmacher P, Schmid E, Hirth L, Knopf B. Carpe diem: A novel approach to select representative days for long-term power system modeling. *Energy* 2016;112:430–42.
- Tejada-Arango DA, Domeshek M, Wogrin S, Centeno E. Enhanced representative days and system states modeling for energy storage investment analysis. *IEEE Trans Power Syst* 2018;33:6534–44.
- Green R, Staffell I, Vasilakos N. Divide and conquer? k-means clustering of demand data allows rapid and accurate simulations of the British electricity system. *IEEE Trans Eng Manag* 2014;61:251–60.
- Poncelet K, Höschle H, Delarue E, Virag A, D'haeseleer W. Selecting representative days for capturing the implications of integrating intermittent renewables in generation expansion planning problems. *IEEE Trans Power Syst* 2017;32:1936–48.
- Fazlollahi S, Bungener SL, Mandel P, Becker G, Maréchal F. Multi-objectives, multi-period optimization of district energy systems: I. Selection of typical operating periods. *Comput Chem Eng* 2014;65:54–66.
- van der Heijde B, Vandermeulen A, Salenbien R, Helsen L. Representative days selection for district energy system optimisation: a solar district heating system with seasonal storage. *Appl Energy* 2019;248:79–94.
- Liu Y, Sioshansi R, Conejo AJ. Hierarchical clustering to find representative operating periods for capacity-expansion modeling. *IEEE Trans Power Syst* 2017;33:3029–39.
- Heuberger CF, Staffell I, Shah N, Mac Dowell N. A systems approach to quantifying the value of power generation and energy storage technologies in future electricity networks. *Comput Chem Eng* 2017;107:247–56.
- Lara CL, Mallapragada DS, Papageorgiou DJ, Venkatesh A, Grossmann IE. Deterministic electric power infrastructure planning: Mixed-integer programming model and nested decomposition algorithm. *Eur J Oper Res* 2018;271:1037–54.
- Manríquez F, Sauma E, Aguado J, de la Torre S, Contreras J. The impact of electric vehicle charging schemes in power system expansion planning. *Appl Energy* 2020;262:114527.
- Pineda S, Morales JM. Chronological time-period clustering for optimal capacity expansion planning with storage. *IEEE Trans Power Syst* 2018;33:7162–70.
- Sun M, Teng F, Zhang X, Strbac G, Pudjianto D. Data-driven representative day selection for investment decisions: a cost-oriented approach. *IEEE Trans Power Syst* 2019.
- Teichgraber H, Brandt AR. Clustering methods to find representative periods for the optimization of energy systems: An initial framework and comparison. *Appl Energy* 2019;239:1283–93.
- Heuberger CF, Rubin ES, Staffell I, Shah N, Mac Dowell N. Power capacity expansion planning considering endogenous technology cost learning. *Appl Energy* 2017;204:831–45.
- De Sisternes FJ, Webster MD. Optimal selection of sample weeks for approximating the net load in generation planning problems. *Massachusetts Inst Technol Eng Syst Div Work Pap Ser CiteSeer* 2013:1–12.
- Scott LJ, Carvalho PMS, Botterud A, Silva CA. Clustering representative days for power systems generation expansion planning: Capturing the effects of variable renewables and energy storage. *Appl Energy* 2019;253: 113603.
- Kotzur L, Markewitz P, Robinius M, Stolten D. Impact of different time series aggregation methods on optimal energy system design. *Renew Energy* 2018;117:474–87.
- Bahl B, Kümpel A, Seele H, Lampe M, Bardow A. Time-series aggregation for synthesis problems by bounding error in the objective function. *Energy* 2017;135:900–12.
- Pfenniger S. Dealing with multiple decades of hourly wind and PV time series in energy models: A comparison of methods to reduce time resolution and the planning implications of inter-annual variability. *Appl Energy* 2017;197:1–13.
- Frew BA, Jacobson MZ. Temporal and spatial tradeoffs in power system modeling with assumptions about storage: An application of the POWER model. *Energy* 2016;117:198–213.
- Hilbers AP, Brayshaw DJ, Gandy A. Importance subsampling: improving power system planning under climate-based uncertainty. *Appl Energy* 2019;251:113114.
- Kohonen T. The self-organizing map. *Proc IEEE* 1990;78:1464–80.
- Pözlbauer G. Survey and comparison of quality measures for self-organizing maps. *Proc. Fifth Work. Data Anal., Vysoké Tatry, Slovakia: Elfa Academic Press* 2004:67–82.
- Liu Y, Weisberg RH, Mooers CNK. Performance evaluation of the self-organizing map for feature extraction. *J Geophys Res Ocean* 2006;111..
- Berndt DJ, Clifford J. Using dynamic time warping to find patterns in time series. *KDD Work.*, vol. 10, Seattle, WA; 1994. p. 359–70.
- Ward JR. Hierarchical grouping to optimize an objective function. *J Am Stat Assoc* 1963;58:236–44.
- Kaufman L, Rousseeuw PJ. Finding groups in data: an introduction to cluster analysis. New Jersey: John Wiley & Sons; 1990.
- Howells M, Rogner H, Strachan N, Heaps C, Huntington H, Kyreos S, et al. OSeMOSYS: the open source energy modeling system: an introduction to its ethos, structure and development. *Energy Policy* 2011;39:5850–70.
- Pena I, Martínez-Anido CB, Hodge B-M. An extended IEEE 118-bus test system with high renewable penetration. *IEEE Trans Power Syst* 2017;33:281–9.
- R Core Team. R: A language and environment for statistical computing; 2019.
- Sarda-Espinosa A. dtwclust: Time series clustering along with optimizations for the dynamic time warping distance; 2019.
- Maechler M, Rousseeuw P, Struyf A, Hubert M, Hornik K. cluster: Cluster Analysis

- Basics and Extensions; 2019.
- [46] Wehrens R, Buydens LMC. Self-and super-organizing maps in R: the Kohonen package. *J Stat Softw* 2007;21:1–19.
 - [47] U.S. Energy Information Administration. Capital Cost Estimates for Utility Scale Electricity Generating Plants. US Dep Energy; 2016. https://www.eia.gov/analysis/studies/powerplants/capitalcost/pdf/capcost_assumption.pdf.
 - [48] U.S. Energy Information Administration. Cost and Performance Characteristics of New Generating Technologies, Annual Energy Outlook 2019. US Dep Energy; 2019. https://www.eia.gov/outlooks/aeo/assumptions/pdf/table_8.2.pdf.
 - [49] Gonzalez-Salazar MA, Kirsten T, Prchlik L. Review of the operational flexibility and emissions of gas-and coal-fired power plants in a future with growing renewables. *Renew Sustain Energy Rev* 2018;82:1497–513.
 - [50] Schröder A, Kunz F, Meiss J, Mendelevitch R, Von Hirschhausen C. Current and prospective costs of electricity generation until 2050. DIW Data Documentation; 2013.
 - [51] Kumar N, Besuner P, Lefton S, Agan D, Hilleman D. Power plant cycling costs. National Renewable Energy Lab.(NREL), Golden, CO (United States); 2012.
 - [52] Elia. Grid data 2019. <https://www.elia.be/en/grid-data> (accessed September 1, 2019).